Arista Access Node HEB-ACC-63 Hebbal Edge Hardware/Environmental Fault Resolution SOP

Node: HEB-ACC-63 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_9260

Description

This Standard Operating Procedure (SOP) addresses a hardware/environmental fault on the Arista Access Node HEB-ACC-63 located at the Hebbal Edge site, resulting in ASIC memory parity errors and line card crashes due to loss of power supply redundancy. The Access Node, being a critical component of the network's edge infrastructure, relies on redundant power supplies to ensure continuous operation. However, in the event of a hardware or environmental failure, the node may malfunction, leading to data plane disruptions and impact on network services. The failure domain for this scenario is rooted in the hardware and environmental fault, which can be triggered by ASIC memory parity errors, line card crashes, or loss of power supply redundancy. The Alarm Code ALARM_9260 is generated to indicate a Minor severity failure. This SOP outlines the diagnostic and resolution steps to rectify the issue and restore network services.

Pre-requisite Checks

- Check: Verify power supply redundancy status using the command 'show power status' to ensure that both power supplies are functioning correctly.

- Check: Confirm the line card status using the command 'show environment status' to ensure that the line card is not crashed.

- Check: Dispatch remote hands to Hebbal Edge site to physically inspect the node and identify any environmental issues.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Loss of power supply redundancy	Low

Diagnostic Commands & Expected Outputs

```
Command: show power status, Expected Output: Power supply 1: OK Power supply 2: OK, Description: This command verifies the power supply redundancy status, ensuring that both power supplies are functioning correctly.

Command: show environment status, Expected Output: Line card 1: Online Line card 2: Online, Description: This command confirms the line card status, ensuring that the line card is not crashed.
```

```
Command: show system internal power, Expected Output: Power supply 1: 220V, 10A Power supply 2: 220V, 10A, Description: This command displays the power supply status, including voltage and current readings, to verify that both power supplies are functioning correctly.
```

Resolution Steps

Step 1: Step: Replace power supply 1, Command: replace power-supply 1, Description: Replace power supply 1 with a new one to ensure power supply redundancy.

Step 2: Step: Reset the line card, Command: reset line-card 1, Description: Reset the line card to restore its original configuration and functionality.

Step 3: Step: Verify power supply redundancy status, Command: show power status, Description: Verify that both power supplies are functioning correctly after replacing power supply 1.

Post-Check Verification

```
Command: show power status, Expected Output: Power supply 1: OK Power supply 2: OK, Description: Verify that both power supplies are functioning correctly.

Command: show environment status, Expected Output: Line card 1: Online Line card 2: Online, Description: Verify that the line card is online and functioning correctly.
```

Roll-back Procedures

- Step: Restore original power supply 1, Command: insert power-supply 1, Description: Restore the original power supply 1 to its previous state.

- Step: Reset the line card, Command: reset line-card 1, Description: Reset the line card to its original configuration and functionality.